GOVERNMENT OF INDIA

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Defence Research & Development Organisation (DRDO)

Centre for Airborne Systems (CABS), Bengaluru – 560 037

TECHNICAL REPORT

AIRBORNE SURVEILLANCE RADAR SYSTEM

MARK I – PERFORMANCE EVALUATION AND FIELD TRIAL ANALYSIS

Report No.: CABS/TR/2024/MK-I/001

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1. INTRODUCTION

The Airborne Surveillance Radar System (ASRS) Mark I is an advanced indigenous airborne Early Warning and Control System (EWACS) developed by the Centre for Airborne Systems (CABS), Bengaluru, under the aegis of the Defence Research & Development Organisation (DRDO). This report presents a comprehensive evaluation of the system performance subsequent to the completion of Phase II field trials conducted over the Rajasthan Test Range during October–November 2024.

The primary objective of the ASRS Mk I programme is to provide the Indian Air Force (IAF) with a state-of-the-art indigenous surveillance capability, thereby reducing dependence on imported systems and contributing to the national mission of 'Atmanirbhar Bharat' in the domain of defence electronics.

2. SCOPE OF THE REPORT

This technical report covers the following aspects of the ASRS Mk I system evaluation:

- System architecture and sub-system integration verification
- Radar performance parameters against defined Key Performance Indicators (KPIs)
- Field trial data analysis over a 30-day operational window
- Reliability, Maintainability, and Availability (RMA) assessment
- Recommendations for Mark II development and system upgradation

3. SYSTEM TECHNICAL SPECIFICATIONS

The following table presents the verified technical parameters of the ASRS Mk I system as measured during the Phase II field evaluation. All values have been independently verified by the DRDO Quality Assurance Directorate (QAD).

Parameter	Specification	Achieved Value
Operating Frequency Band	L-Band (1–2 GHz)	1.35 GHz (Centre)
Detection Range (Air Target)	400 km (min.)	500 km
Altitude Coverage	100 m – 15,000 m	50 m – 15,500 m
Number of Simultaneous Tracks	150 (minimum)	200 tracks
Data Link Range	300 km	350 km
Radar Rotation Rate	6 RPM	6 RPM
Transmitter Peak Power	1.5 MW	1.52 MW
Pulse Width	10 μs / 50 μs	10 μs / 50 μs
MTBF	500 hours (min.)	580 hours
Operating Temperature	-20°C to +55°C	-25°C to +58°C

4. PERFORMANCE METRICS & OBSERVATIONS

Figure 1: Radar Coverage Map – Detection Range 500 km (Phase II Trials)

4.1 Field Trial Performance Summary

Field trials conducted at the Rajasthan Test Range yielded data across 847 sorties over a 30-day evaluation window. The ASRS Mk I demonstrated performance exceeding the defined KPIs across all major parameters. Key observations are summarised as follows:

- Detection probability (Pd) of 0.94 achieved against RCS = 1 m² targets at 450 km range
- False Alarm Rate (FAR) maintained below 10^{-6} per scan during all trial phases
- Electronic Counter-Countermeasures (ECCM) validated against simulated jamming scenarios
- Identification Friend or Foe (IFF) Mode S integration confirmed operational
- System availability achieved at 97.3% over the 30-day trial window
- Data link demonstrated at 350 km range with end-to-end latency below 500 ms

Figure 2: Satellite Data Link Orbital Configuration for ASRS Mk I Integration

4.2 Competitive Comparison Table

System Parameter	ASRS Mk I (DRDO)	Baseline Req.	Status
Detection Range	500 km	400 km	**EXCEEDED**
Track Capacity	200 tracks	150 tracks	**EXCEEDED**
MTBF	580 hrs	500 hrs	**EXCEEDED**
Jamming Resistance	78 dB/MHz	70 dB/MHz	**EXCEEDED**
Power Consumption	185 kW	200 kW (max)	**MET**
System Weight	3,420 kg	3,500 kg (max)	**MET**

5. ANALYSIS

5.1 Radar Sub-System Analysis

The radar sub-system demonstrated superior performance across all specified test scenarios. The phased array antenna achieved a beam-forming accuracy of ±0.05° throughout the temperature range, significantly exceeding the ±0.15° specification. Signal processing latency remained below 8 ms for all target types evaluated, enabling real-time tracking of fast-manoeuvring targets at ranges up to 480 km.

The ECCM suite, comprising frequency agility, pulse compression, and adaptive sidelobe cancellation, was validated against advanced jamming platforms provided by the Electronic Warfare (EW) Wing. The system successfully maintained track on 85% of targets even under heavy barrage jamming conditions at a 78 dB/MHz jamming-to-signal ratio.

5.2 Data Link and Communications Analysis

The tactical data link demonstrated consistent performance at ranges up to 350 km, utilising the Link 16-compatible waveform developed indigenously by DRDO-DLRL, Hyderabad. Throughput of 1.12 Mbps was consistently achieved during ground and airborne test phases, sufficient to support 200 simultaneous track updates per second.

5.3 Reliability and Maintainability

A total of 14 minor corrective maintenance actions were recorded over the 30-day trial period. No major system failures requiring depot-level maintenance were encountered. The Mean Time to Repair (MTTR) averaged 2.4 hours, well within the 4-hour specification. The Built-In Test Equipment (BITE) detected 92% of fault conditions automatically, reducing dependence on manual fault isolation procedures.

Figure 3: System Performance Analysis – ASRS Mk I (Phase II Field Trials)

6. CONCLUSION

The ASRS Mk I system has successfully completed Phase II field trials with all Key Performance Indicators met or exceeded. The system demonstrated a detection range of 500 km, track capacity of 200 simultaneous targets, and an MTBF of 580 hours – representing a significant advancement in indigenous airborne surveillance capability. The overall system availability of 97.3% confirms suitability for operational deployment with the Indian Air Force.

The technology demonstrator phase has conclusively validated all critical sub-systems including the indigenous AESA radar, signal processor, data link, and mission computer. The ASRS Mk I represents a landmark achievement in DRDO's pursuit of self-reliance in advanced defence electronics.

7. RECOMMENDATIONS

- Immediate initiation of Limited Series Production (LSP) clearance process for 3 ASRS Mk I platforms
- Allocation of additional development funding for ASRS Mk II incorporating AESA technology and extended detection range beyond 700 km
- Establishment of a dedicated MRO facility at Air Force Station Lohegaon for ASRS platform support
- Integration of Advanced IFF Mode 5 in the production configuration
- Commencement of joint trials with the Navy for maritime surveillance variant development
- Formation of a Technology Transfer Working Group with HAL for indigenous production scaling

DOCUMENT APPROVAL

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